

Lecture 16: Exponential function.

Question: Is there function $f: \mathbb{R} \rightarrow \mathbb{R}$ s.t.

$$\boxed{f' = f} \quad (f(0) = 1)$$

Let's try it out:

$$f_0 = 1 \quad f'_0 = 0.$$

$$f_1 = x + 1 \quad f'_1 = 1$$

$$f_2 = \frac{x^2}{2} + x + 1 \quad f'_2 = x + 1$$

$$f_n = \frac{x^n}{n!} + \frac{x^{n-1}}{(n-1)!} + \dots + 1 \quad \underline{f'_n = f_{n-1}}$$

Are we getting anywhere with this?

Yes, kind of: For $x_0 \in \mathbb{R}$

$$f_{n+1}(x_0) - f_n(x_0) = \frac{x_0^{n+1}}{(n+1)!} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$$f_{n+1}(x_0)$$

→ The f_n 's almost satisfy the eq. $\overline{f' = f.} \quad (*)$

We want to define a limit of the functions f_n , because then we may hope that this limit satisfies $(*)$.

In other words: We want a function

$$f(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\left[\begin{array}{l} \text{In an ideal world we even have} \\ f'(x) = \sum_{k=0}^{\infty} \left(\frac{x^k}{k!} \right)' = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = f. \end{array} \right]$$

How can we make this precise?

1) If we can show that $\forall x \in \mathbb{R}$
the seq. $f_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$ converges

to some numbers $\left(\sum_{k=0}^{\infty} \frac{x^k}{k!} \right)$

then we can define a function

$$f(x) = \left(\lim_{n \rightarrow \infty} f_n(x) \right) = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

• We say f is the pointwise
limit of the functions f_n .

(we also say $f_n \xrightarrow{\text{pt.}} f$).

[2] Is it true that

$$\frac{d}{dx} \left(\lim_{n \rightarrow \infty} f_n \right) = \lim_{n \rightarrow \infty} \left(\frac{d}{dx} f_n \right).$$

Lemma 1 let $x \in \mathbb{R}$. Then the series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges (this means $\sum_{k=0}^n \frac{x^k}{k!}$ converges as $n \rightarrow \infty$)

(remember: $\sum_{k=0}^{\infty} q^k$ converges if $0 \leq q < 1$).

General criterion: for a series $\sum_{k=0}^{\infty} a_k$ to converge: $\exists N \exists r \in [0, 1)$ st.

$$\forall n \geq N \quad |a_{n+1}| \leq r \cdot |a_n|$$

In our case: $a_{n+1} = \frac{x^{n+1}}{(n+1)!} = \frac{x}{(n+1)} \cdot a_n$

$$a_n = \frac{x^n}{n!}$$

If $n \gg 0$ then $\frac{x}{n+1} < 1$.

Pick N s.t. $\frac{x}{N+1} < 1$ then

the ratio criterion is satisfied
with $r = \frac{x}{N+1}$.

Defn. We define the function

$\exp: \mathbb{R} \rightarrow \mathbb{R}$ as

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad (= \lim_{n \rightarrow \infty} f_n(x))$$

What about the diff. of $\exp$?

Assume we have a seq. of fct.

$$f_n: \mathbb{R} \rightarrow \mathbb{R}$$

$\forall x:$

s.t. $\left[\begin{array}{l} \bullet \exists f: \mathbb{R} \rightarrow \mathbb{R} \text{ s.t. } f(x) = \lim f_n(x), \\ \bullet \forall n \ f_n' \text{ is diff!} \end{array} \right.$

$$\exists g: \mathbb{R} \rightarrow \mathbb{R} \text{ s.t. } g(x) = \lim_{n \rightarrow \infty} f_n'(x) \quad \forall x \in \mathbb{R}.$$

We may try to prove that f is diff at some $x \in \mathbb{R}$.

For given $\varepsilon > 0$ we find $\delta > 0$ s.t. $\forall x' \in (x - \delta, x + \delta)$.

$$\left| \frac{f(x') - f(x)}{x' - x} - g(x) \right| < \varepsilon$$

$$\left| \underbrace{\frac{f(x') - f(x)}{x' - x} - \frac{f_n(x') - f_n(x)}{x' - x}}_{1)} + \underbrace{\frac{f_n(x') - f_n(x)}{x' - x} - g(x)}_{2)} \right|$$

Goal: to control

$$1) \left| \frac{f(x') - f(x)}{x' - x} - \frac{f_n(x') - f_n(x)}{x' - x} \right|$$

$$2) \left| \frac{f_n(x') - f_n(x)}{x' - x} - g(x) \right|$$

$$\forall x', x_0 \in (x - \delta, x + \delta) \quad (*)_2.$$

Problem With our assumptions
this is impossible!

We need to control

$$|f_n - f| \text{ and } |f_n' - g|$$

$\forall x$ in a given interval!

Defn We say a seq of functions f_n converges absolutely to f on $S \subseteq \mathbb{R}$ if $\forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \geq N$ s.t. $\forall x \in S: |f_n(x) - f(x)| < \epsilon$.

Theorem: In the situation above
(*) If $f_n' \rightarrow g$ absolutely on an interval (a, b) then $\forall x \in (a, b)$

$\lim_{n \rightarrow \infty} f_n = f$ is diff'ble
at x

and $f'(x) = g(x)$.

Pf. Continue with the computation
above. We need to find

N and δ st. $\forall x' \in (x-\delta, x+\delta)$

$$1) \left| \underbrace{\frac{f(x') - f(x)}{x' - x}}_Q - \underbrace{\frac{f_n(x') - f_n(x)}{x' - x}}_{Q_n} \right| < \varepsilon$$

We need to show that

$Q_n \rightarrow Q$ absolutely.

Trick: If $Q_n \rightarrow Q$ conv. pointwise
 & Q_n is absolutely Cauchy
 then $Q_n \rightarrow Q$ absolutely.

Indeed: $|Q_n^{(x')} - Q_n^{(x)}|$

$$= \left| \frac{f_n(x') - f(x)}{x' - x} - \frac{f_n(x) - f(x)}{x' - x} \right|$$

$$= \left| f_n'(x_1) - f_n'(x_2) \right|$$

↗ becomes arbitrarily small

for n, n large!
 (b.c. f_n' is abs. convergent).

$$2) \left| \frac{f_n(x') - f_n(x)}{x' - x} - f'(x) \right|$$

$$= \left| \underbrace{f_n(x') - f_n(x)}_{\substack{\downarrow \\ 0 \text{ as } n \rightarrow \infty \\ \text{since } f_n \text{ is diff.}}} - f_n'(x) + \underbrace{f_n'(x) - g(x)}_{\substack{\downarrow \\ 0 \text{ as } n \rightarrow \infty \\ \text{b.c. } f_n' \rightarrow g.}} \right|$$

Upshot: Apply this to exp!

We have to show that

$$f_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$$

converges to exp uniformly

on any interval of the

form $(-r, r)$ for $r \in \mathbb{R}$.

Pf. 1st that the f_n 's are uniformly
Cauchy on $(-r, r)$.

For this we only the following
observation: ($n > m$)

$$|f_n(x) - f_m(x)| \leq \left| \sum_{k=m+1}^n \frac{x^k}{k!} \right|$$

$$\leq \sum_{k=m+1}^n \frac{r^k}{k!}$$

get small
if $n, m \gg 0$

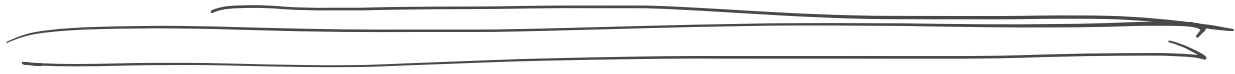
$\Rightarrow$ f_n 's are un. uniformly

Cauchy

$$\Rightarrow \exp'(x) = \lim_{n \rightarrow \infty} f_n'(x)$$

$$= \exp(x)$$

$\forall x.$



Complex numbers

Idea. Introduce a new number i ,
satisfying $i^2 = -1$.

$$\mathbb{C} = \{ x + iy \mid x, y \in \mathbb{R} \}.$$

Addition & Multiplication of the
real numbers extend naturally

$$\text{to } \mathbb{C} \quad (\mathbb{R} \subseteq \mathbb{C})$$

$$x \mapsto x + i \cdot 0$$

Notation: $z = x + iy \in \mathbb{C}$

$$\bullet \operatorname{Re}(z) = x \in \mathbb{R}$$

$$\bullet \operatorname{Im}(z) = y \in \mathbb{R}.$$

Fact 1: $\mathbb{C}$ is a field.

(i.e. every $z \in \mathbb{C} \setminus \{0\}$ has a

mult. inverse: $z = x + iy$

$$z \cdot \frac{(x - iy)}{x^2 + y^2} = \frac{x^2 - i^2 y^2}{x^2 + y^2} = 1.$$

Notation: $x - iy = \bar{z}$ (Complex
conj. of z)

Fact 2. There is a abs. value

function $|\cdot|: \mathbb{C} \rightarrow \mathbb{R}$

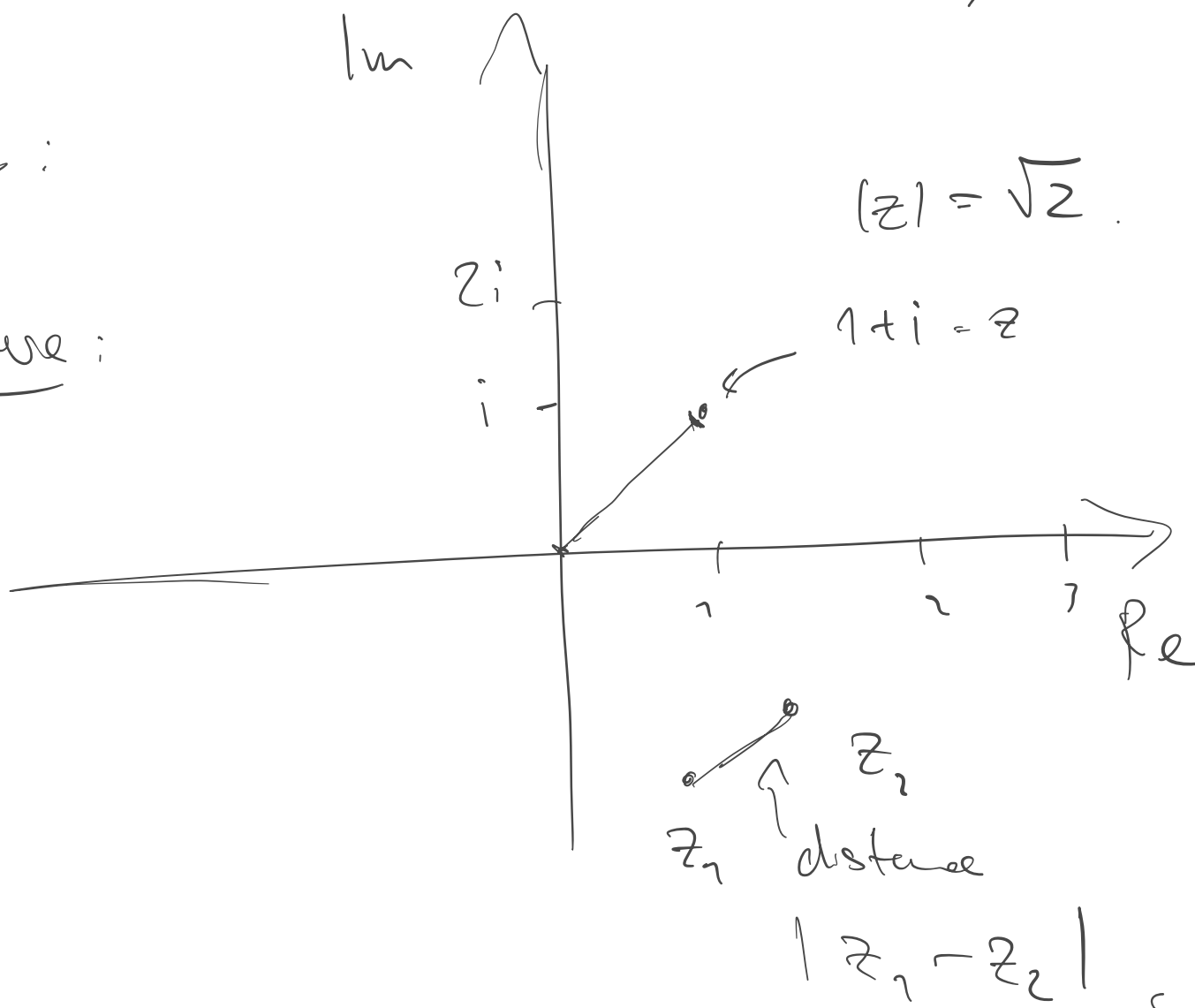
$$\bullet |z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

$$\bullet |z_1 + z_2| \leq |z_1| + |z_2|$$

$$z \mapsto \sqrt{z \cdot \bar{z}} \\ = (x^2 + y^2)$$

$\mathbb{C}$:

Picture:



→ This induces a notion of
distance on $\mathbb{C}$

$$(\text{dist } z_1 z_2 = |z_1 - z_2|)$$

→ All notions of convergence
that we talked about
in the course carry over
1-1.

Directions to go into:

• Fundamental theorem of algebra:

→ Every complex polynomial has a root.

• We can work out a theory
of complex analysis. In part,

Complex derivation.

For example: $f' = \lim_{z' \rightarrow z} \frac{f(z') - f(z)}{z' - z}$.

→ This leads rich theory of complex analytic functions.

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- 1) General notions of convergence
 - 2) Cauchy-Riemann equations for complex analytic f .
 - 3) Cauchy-Int. formula ...

- Power series in complex variables
(exp, sine, cosine).

By studying the complex exp
function (& sine / cosine) you

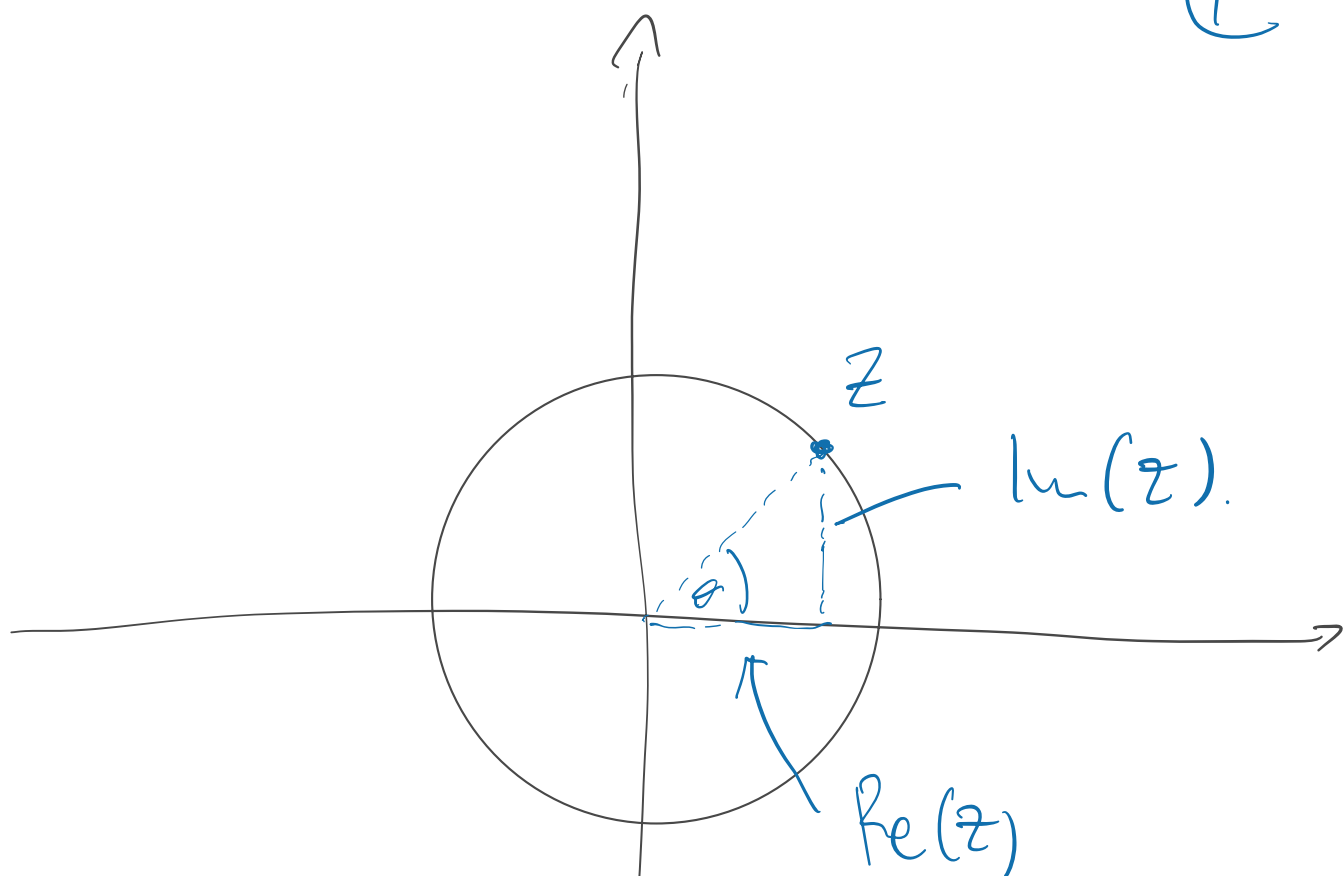
can prove:

- Euler identity $\exp(i\pi) = -1$.

$$\left[\begin{array}{l} \exp(i\theta) = \cos\theta + i\sin\theta \\ \text{for } \theta \in \mathbb{R}. \end{array} \right]$$

- Working harder leads to
every $z \in \mathbb{C}$ is the form
 $z = r \cdot \exp(i\theta)$ for $r \in \mathbb{R}_{\geq 0}$.

①



$$\begin{aligned}
 z &= \operatorname{Re} z + i \ln(z) \\
 &= \cos \theta + i \sin \theta \\
 &= \exp(i\theta).
 \end{aligned}$$

You can also deduce this purely from the power series defn's of $\sin$ & $\cos$.

And the consequences of Euler's identity : You can compute

$$(\cos \theta + i \sin \theta)^n = \exp(i\theta)^n$$

$$= \exp(in\theta)$$

$$= \cos n\theta + i \sin n\theta$$